



Securing Wireless Sensor Network Against Multi-Malware Threats: A Comprehensive Modeling and Analysis Approach

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Abstract

Wireless sensor networks (WSNs) are composed of disparately positioned sensor nodes that collaborate to monitor and identify alterations in the surrounding environment for a variety of uses, comprising commercial, military, and residential domains. Due to limited resources, sensor nodes are susceptible to malware attacks from Trojan horses, viruses, worms, and dangerous signals. This paper addresses the critical issue of protecting WSNs from malware attacks, especially in light of the potential for multiple malware kinds to coexist, which presents a severe risk to WSN stability. The study looks at two types of malware infections at the same time. To investigate malware transmission in WSN. Caputo's fractional derivative of order $0 < \alpha \leq 1$ is used to analyze the SE_1E_2VIQR epidemic model. The study initially uses the next-generation technique to determine the model's fundamental reproduction number, which is essential for finding the malware-free system's locally asymptotically stable equilibrium point and its unique endemic equilibrium. The described states allow early identification of malware occurrences in WSNs. We numerically solve the model using the Adam–Bashforth–Moulton (ABM) predictor–corrector approach. This method is ideal for resolving the “memory effect” in many phenomena, where system behavior depends on past states. The fractional calculus framework allows this memory effect, revealing the system's complicated dynamics. The findings show that vaccination and quarantine strategy effectively stop WSN malware spread. The computational analysis reveals that the ABM method is robust and efficient for investigating fractional-order dynamical systems, especially the SE_1E_2VIQR malware propagation model. The study also shows that fractional derivative order strongly affects system dynamics. When $\alpha = 1$, the system exhibits integer-order dynamics without the memory effect. However, when α declines from 1, the memory effect intensifies, affecting system behavior over time. This contrasts fractional and integer-order derivatives. The suggested model's theoretical basis are mathematically proven. The model's ability to understand and mitigate WSN malware spread is supported by extensive theoretical study and computational tests.

Keywords Reproduction number · Malware · Exposed state · Equilibrium · Wireless sensor network

Introduction

Wireless sensor networks (WSN) is a wireless network containing a large number of infrastructure-less sensor nodes deployed in the region of interest. The WSN are very

much helpful in monitoring the environment related events as humidity, earthquake, disaster recovery or biological detection, military surveillances, etc. [1–3]. The WSN has a very wide range of application in homes, hospitals, factories and in industrial fields. The WSN gives the real time information also in very harsh situations and unreachable places. These sensors communicate wirelessly and are often used in applications where wired networks are impractical or impossible to implement [4, 5]. These wireless networks typically contain individual sensor nodes, which collect data, and a central node, which processes and manages the data collected by these individual sensor nodes. The sensor nodes are important components of WSN as these are

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responsible for the accuracy and performance of WSN [6]. These sensor nodes are very tiny, budget fitted and are self-organized in nature. Sensor nodes are multi-functional. An individual sensor node contains a sensor according to the functioning of WSN which senses the particular region of interest, a controller which collect the relevant data and store it, a transmitter and a receiver for the well communication between the neighbouring sensor nodes and a power supply. It may contain other components also according to the need of WSN. These sensor nodes are application based and task specific. The use of fire detectors in wild areas is a common example of WSN as it contains sensor nodes for sensing the flame, humidity, smoke, light intensity etc., and then these sensor nodes will transmit the required data to the base station of the WSN which will forward the computed data to network interface for taking the required action [1].

The main functioning of the sensor nodes is to sense the environment according to the need of WSN and deliver the collected data to their neighbour and the receiving node forward the data to sink node by transferring the data to their neighbours [6, 7]. Sink node is used to collect data from different sensor nodes and give ease to communicate the user and network with each other. The collected data are routed to the sink through a single-hop or multi-hop networking. If the sink is at a far distant from the sensor node, then a multi-hop technique is used to transfer the data towards the sink [8].

With the variety of application of WSN, it comes with some challenges also as low power supply, less mobility range, less computation problem, low data storage capacity, threat of data confidentiality, quality of data transferring services etc. The sensor nodes are limited in energy, computational ability and in wireless communication range [9, 10].

Since a wireless sensor network also faces some challenges regarding the security threat as it has a multi-hop networking with open wireless medium and behaviour of seeking support from neighbouring nodes, makes it a very easy target for attackers. But the main objective of the WSN is to provide an authenticate and confidential data [8, 11].

It becomes very difficult to design a very secure and protected sensor network. As sensor nodes are densely deployed in ambient situations with limited and irreplaceable power supply, lack of resources and weak defence ability introduces them to malicious software. This malicious software also known as malware, are software program or code which tries to either extract important information secretly from sensor nodes or temper the nodes. These malwares are designed to disrupt, damage or gain unauthorized access to WSN, or data, often without the user's knowledge. These kinds of malicious activities

can be easily detected by Intrusion Detection Systems (IDS). These malware enters in WSN in form of malicious sensor nodes and tries to replace the genuine sensor nodes, misguide genuine sensor nodes, destroy data packets, replicate malicious sensor nodes, harm the confidentiality, etc. Worms, viruses, ransomware, etc., are some of the types of malware. Once malware enters in the network, it may make an effort to propagate further by scanning vulnerable systems. Propagation of malware means the spread of malicious software in the WSN [12]. To reduce the propagation of malware in the network, user should employ security. Therefore, to understand the model of WSN and to secure our WSN from these kinds of worms, it is important to understand the method of propagation of these worms. These infected nodes also effect the neighbouring nodes by propagation of malware as in the case of disease an infectious person also infect others with the direct person to person contact. This kind of behaviour is similar to the propagation of infectious disease in a region, which can be studied by epidemiological model of disease transformation in living beings [13, 14]. Epidemiological models are tools for estimating disease trends, assessing the potential impact of interventions, and informing public health policies. Epidemiology is the branch of science which reveal the incidence, distribution, control of diseases. It mainly concerns on the pattern and cause of diseases in the population so that the damages can be repaired. Epidemiological modelling consists the compartmental distribution of the different stages of a disease such as susceptible, infectious, exposed, etc. The simplest epidemiological modelling is susceptible-infected-removed (SIR) model used for worm propagation on internet. Numerous academics have already used epidemiological modeling to model the spread of malware in wireless sensor networks [15, 16].

An SIR model for analyzing the dynamics of multi-worm propagation in WSN is presented in [12]. This model considers the sleep and work interleaving schedule policy for sensor nodes and shows that WSN is sensitive towards the energy consumption of nodes. In the paper [15], SEIRS model is discussed on the spreading of virus. The model considers the strongly correlated and less correlated nodes for analysing the dynamics of virus spreading in wireless system. The model specifies the malicious code propagation based on the size or density of the considered epidemiological compartment. In [13], has an SIIDR model for the self-propagating malware. Kermack and McKendrick's epidemiological mathematical model [17] has served as the foundation for several mathematical models in this area. We have mathematically investigated the evolution of an epidemic in a homogeneous population [17]. By emphasizing the attacking behavior of worms [18], the paper builds a SIRS epidemic model. The paper

investigates the stability of worm prevalence by establishing the equilibrium of the model and considering various elements such as energy consumption, communication radius, and distributed density of nodes in WSN. This paper [19] presents an expansion of the SIR model by incorporating the novel concept of a dead state for nodes in wireless sensor networks (WSNs). The research [19] delves into the process of worm propagation, emphasizing the energy consumption of nodes and multi-worm propagation in Wireless Sensor Networks (WSN), and asserts that network architecture and node energy consumption in WSNs influence worm propagation. The research [20] delineates a SE_1E_2IR model featuring two distinct exposed types of worms transmitted in Wireless Sensor Networks (WSN). The research primarily examines a model featuring two distinct latent periods: one characterized by a short latent period, which induces nodes to enter sleep mode, and the other defined by a long latent period, where nodes remain in an active state. The subsequent paper [21] is an updated version of the preceding work [20]. This work [21] presents the SE_1E_2IR model, which claims that prioritizing the prevention of short-latent period malware is more critical than that of long-latent period malware. Short-latent period malware disseminates rapidly and effortlessly, resulting in a greater quantity of infected nodes and accelerated network deterioration. Authors have examined the influence of communication radius and distributed node density on malware propagation.

This is well established that the fractional derivatives and fractional integral have non-local properties [22–24] i.e. it can present both the history and the current effect of any physical systems. This shows the great ability of fractional calculus to represent complicated real-world problems more effectively than ordinary calculus. Due to this specific advantage of fractional calculus many researches of various fields of basic sciences and Engineering used it drastically to model the memory based problems [25–41]. Since signal transmissions are known as memory and hereditary processes [42–44] that often significantly depend on the flexibility of the environment, the history of functions and characteristic properties of the medium, that are tough to describe accurately by means of integer order differential systems. This forces us to apply the fractional calculus in the modelling of propagation of malware in WSN.

Some Related Work

Some recent work who used fractional calculus in this field:

The fractional SE_1E_2IQR epidemic model, with latent periods and quarantine in mean-field reaction rate

equations, is proposed in paper (45) for analyzing malware in heterogeneous networks. The paper (46) introduces a SEIVR model and numerically solves it using the Adam-Bashforth-Moulton predictor–corrector (ABM) method. The study’s findings show that the ABM predictor–corrector method is a strong and effective way to look at fractional order dynamical systems using the SEIVR epidemic model to see how malware spreads in wireless sensor networks. Tahir et. al. [47] discussed Dynamic properties of the multi-malware attacks in wireless sensor networks using Fractional derivative in wireless sensor networks. They use fractional SE_1E_2IR model to explain worm propagation in WSNs.

In this paper we have used the epidemiological compartmental model SE_1E_2VIQR modelling for multi-malware propagation in WSN. In the model, each compartment has its own functioning as S stands for Susceptible compartment, E_1 and E_2 are Exposed, V for Vaccination, I means Infectious, Q means Quarantine, R means Recovered. The paper is structured as follows: Sect. [Some Related Work](#) gives definitions and results used throughout the paper. Section [Fundamental Definition and Outcome](#) explains the model in detail, addressing its assumptions. Section [Model Explanation and Presumptions Used in Its Analysis](#) discusses the motivation behind the proposal of this model. Section [Motivation](#) presents the model’s formulation and an overview of the parameters. Section [Formulation of the Model](#) looks at whether the epidemic model has a nonnegative solution and whether it is unique. In Sect. [Description of Uniqueness and Existence of Non-Negative Solution of the Model](#), we talk about the Basic Reproduction Number $\mathcal{R}_0$ and look at the stability of both the malware equilibrium and the endemic equilibrium of the epidemic model. Finally, Sect. [Analyzing the Model’s Equilibrium and Stability](#) assesses the model’s accuracy and shows numerical results through graphical representations.

Fundamental Definition and Outcome

Initially, we review some fundamental definitions and findings related to fractional calculus.

Definition 1. Let a function f define on $[a, b]$. The Riemann–Liouville fractional integral of order $\alpha > 0$ for function f is defined as:

$${}_a^{RL}D_x^{-\alpha}f(x) = \frac{1}{\Gamma(\alpha)} \int_a^x (x-s)^{\alpha-1}f(s)ds, x \in [a, b].$$

We obtain ${}^RLD_x^\alpha = I$ for $\alpha = 0$, which is the identity operator.

Definition 2. Let $\alpha \geq 0$, $n \in \mathbb{N}$ and $f \in C^n[a, b]$. For the function f , the Caputo fractional derivatives of order α is defined by.

$${}_a^c D_t^\alpha f(t) = \frac{1}{\Gamma(1-\alpha)} \int_a^t \frac{f'(s)}{(t-s)^\alpha} ds, t \in [a, b], 0 < \alpha \leq 1$$

Definition 3. Let $\beta > 0$, the function $\mathcal{E}_\beta$ defined by.

$$\mathcal{E}_\beta(z) = \sum_{n=0}^{\infty} \frac{z^n}{\Gamma(n\beta + 1)}, \beta > 0, z \in \mathbb{C}$$

Is referred to as the α order Mittag-Leffler function $\mathcal{E}_\beta(z)$ is an entire function. When $\beta = 1$, $\mathcal{E}_1(z) = e^z$ and when $\beta = 2$, $\mathcal{E}_2(z) = \cosh(\sqrt{z})$ are the particular circumstances. The standard form is

$$\mathcal{E}_{\alpha, \beta}(z) = \sum_{n=0}^{\infty} \frac{z^n}{\Gamma(n\alpha + \beta)}, \alpha, \beta \in \mathbb{C}, z \in \mathbb{C} \text{ and } R(\alpha) > 0, R(\beta) > 0$$

Lemma 1. If $\alpha \geq 0$ and f is continuous, then ${}_a^c D_t^\alpha {}_a^c D_t^\alpha f(t) = f(t)$.

Lemma 2. If $r > 0, \theta \in [-\pi, \pi], \alpha \geq 0$ and $\lambda = re^{-i\theta}$, then $\lim_{t \rightarrow \infty} \mathcal{E}_\alpha(-\lambda t^\alpha) = 0$ for $|\theta| < \frac{\pi}{2}$.

Here are some findings regarding the fractional order systems' stability.

Assume that D is an open subset of $\mathbb{R}^n$. Consider the initial value problem (IVP) For $\alpha \in (0, 1]$, which is composed of an autonomous fractional-order system.

$${}_a^c D_t^\alpha u(t) = h(u), u(t_0) = u_0 \quad (2.1)$$

where $h : D \rightarrow \mathbb{R}^n$ is locally Lipschitz continuous in t .

Definition 4. The system (2.1) is said to be in equilibrium at point u^* if $h(u^*) = 0$.

Lemma 3. [22, 48] Let $J(u^*)$ denotes the Jacobian matrix of the system calculated at an equilibrium point u^* , and let $\lambda_j, j = 1, 2, 3, \dots, n$ be the eigen values of $J(u^*)$. Then u^* is locally asymptotically stable if and only if $|\arg \lambda_j| > \frac{\alpha\pi}{2}, j = 1, 2, 3, \dots, n$.

To prove stability, we have the following lemma:

Lemma 4. Suppose for any $a \geq 0, g \in C([a, \infty), \mathbb{R}^+)$ and $\alpha \in (0, 1]$ For any $a \geq 0$, we have.

$${}_a^c D_t^\alpha g^2(t) \leq g(t) {}_a^c D_t^\alpha g(t) \text{ [ref.[48], Lemma 1].}$$

$${}_a^c D_t^\alpha \left[g(t) - g^* - g^* \ln \frac{g(t)}{g^*} \right] \leq \left(1 - \frac{g^*}{g(t)} \right) {}_a^c D_t^\alpha g(t),$$

where $g^* \in \mathbb{R}^+$.

Let $H \in C^1(D, \mathbb{R})$ and $\alpha \in (0, 1]$, the α^{th} order Caputo derivatives of $H(u)$ along the solution $u(t)$ of the system ${}_a^c D_t^\alpha u(t) = h(u), t \in [a, \infty)$ is given by [24]

$${}_a^c D_t^\alpha H(u(t)) = {}_a^c D_t^{-(1-\alpha)} \left(\frac{dH}{du} \frac{du}{dt} \right)$$

Next is the fractional representation of the widely recognized LaSalle's invariance Principle:

Lemma 5. [48] Let any solution of ${}_a^c D_t^\alpha u(t) = h(u), t \in [a, \infty)$, beginning at a point in A , stays in A for all time t . Here bounded closed sets in $\mathbb{R}^n$ are assumed to be A .

Furthermore, assume that $H \in C^1(A, \mathbb{R})$ with ${}_a^c D_t^\alpha H(u(t)) \leq 0$, where $u(t)$ is any solution of the system ${}_a^c D_t^\alpha u(t) = h(u)$. Let S be the largest invariant subset of E and let $E = \{u \in A : {}_a^c D_t^\alpha H = 0\}$. Then, as $t \rightarrow \infty$ every solution $u(t)$ of ${}_a^c D_t^\alpha u(t) = h(u), t \in [a, \infty)$; starting in A tends to S . To be more precise, $u(t) \rightarrow 0$ as $t \rightarrow \infty$ if $S = \{0\}$.

Model Explanation and Presumptions Used in Its Analysis

To protect WSN against worm attacks, a defense mechanism is required. The SE₁E₂VIQR (Susceptible-Exposed-Infectious-Quarantined-Vaccinated-Recovered) fractional epidemic model is suggested for this reason. The formulation of the model, which is shown in Fig. 1, aids in the study of the dynamics of worm propagation in WSN. Infectious states of the sensor nodes are divided into two categories in the model. Infectious state sensor nodes and exposed state sensor nodes are the two categories. Worms exploit nearby active mode sensor nodes to communicate valuable data in a wireless sensor network.

The different states of the proposed models:

Susceptible State(S): These are nodes that are vulnerable to worms but are not infected.

Exposed State (E₁ and E₂): These are non-infectious but infected nodes. Nodes of this type do not spread to neighboring nodes.

Infectious State(I): These nodes are contaminated and can infect further nodes.

Quarantine State(Q): Quarantine nodes are those nodes which are isolated from other nodes interaction.

Recovered State(R): A node is said to be in a recovered state if it was previously infected but is now free of malware.

Vaccinated State(V): The vaccinated state is believed to consist of those nodes that have received vaccinations.

Motivation

The basic motive for development of the model is to prevent malware transmission and increase WSN's lifespan. The model explains the dynamics of malware transmission process in WSN when two types of malwares present in the network, the significant contributions of proposed model is:

1. To investigate the dynamics of malware transmission process in WSN when two types of malwares appear in the network simultaneously. The model analyses the transmission behaviour of malwares and suggests the mechanism for controlling of malwares transmission in WSN.
2. To utilize the concept of different latent period to protect the network against malwares attack. On the basis of latent period of different malwares apply the corrective measures to safeguard WSN. The suggested mechanism helps in securing and enhancing the lifetime of WSN.
3. To stop malware attacks and save energy consumption of sensor nodes in WSN by recovering vulnerable sensor nodes.
4. To study the impact of spatial as well as temporal parameters on transmission of malware in WSN.
5. To examine the system's stability under various conditions and verify analytical findings with the help of comprehensive simulation results.

Some of the assumptions are made for formulation of the model. The identical type of sensor nodes is distributed in a field to gather the information from the environment and transfer to the destination/sink node/control centre in single hop or multi-hop. All sensor nodes are of in Susceptible (S) state in the beginning and free from malicious signals. A sensor node of WSN is targeted by the attacker and installs malware. When the malware is successfully installed on the network and begins communicating with other nodes on the network, the malware attack changes the state of the sensor node.

Formulation of the Model

The following presumptions were made in order to formulate the model:

- At first, sensor nodes are in the Susceptible state (S), which leaves them open to malware attacks. A vulnerable sensor node turns into an exposed node when

malware infiltrates it. On an exposed node, malware is partially executed. The proposed model consider two types of malware attack at the same time, which are categorised by latent period as first with short latent period and other with large. The rates at which susceptible nodes become exposed nodes are $a\beta SI$ and $b\beta SI$ (where $a + b = 1$). Normal functioning causes some vulnerable sensor nodes to die at a rate of μ .

- Apply vaccination to susceptible nodes to prevent them from infection. So, some of the susceptible nodes moves to Vaccinated state with rate of θS . But these vaccination gives temporary immunity to the sensor nodes so these vaccinated sensor nodes again become exposed to infection with the given two categories of exposed E_1 and E_2 at the rate of $\sigma_1 \beta VI$ and $\sigma_2 \beta VI$. Some of these vaccinated nodes may be demolished due to deficiency of energy.
- In a wireless sensor network, the exposed state (E) of a sensor node indicates aberrant behaviour. Malware has installed and partially executed on the vulnerable node. As a result, implement a corrective action on exposed nodes in a timely manner to prevent the spread of malware in WSN. If not then these nodes may become infectious at a rate of $\alpha_1 E_1, \alpha_2 E_2$, respectively. A malware assault causes a portion of the exposed class of sensor nodes to die at a rate of μ .
- Malware compromises the sensor nodes of WSN, allowing malicious activity to start within the network. Infected nodes increase their own energy dissipation and begin to propagate malware among nearby nodes. Malicious programs may also choose to stop targeting the compromised node; infected nodes are only contagious and not harmful. When the remedial action is implemented on time and anti-malware is successfully installed, then the infected sensor nodes move to the isolated state known as quarantined state with a rate of γI . A portion of the sensor nodes in quarantine may perish at a rate of μ .
- At a rate of δQ , the nodes will now be safely transformed to the recovered state. This means that the node will die faster as the amount of damage caused by malicious applications increases. Worm attacks cause some of the infecting sensor nodes to die at a rate of μ . Nodes in the recovered state of a wireless sensor network (WSN) possess high energy levels and are resistant to malware attacks. The network's sensor nodes enter a state of recovery. Some of the recovered class of sensor nodes demise at a rate of μ .
- The system is adding susceptible sensor nodes at a rate of Λ (Table 1).

At any time, t all the sensor nodes of WSN are divided into seven states. The system has $N(t)$ nodes with the

following names: Susceptible State $S(t)$; Vaccinated $V(t)$; Exposed State $E(t)$; Infectious State $I(t)$; Quarantined State $Q(t)$; Recovered State $R(t)$.

Therefore, mathematically satisfy the following equation $N(t) = S(t) + E_1(t) + E_2(t) + V(t) + I(t) + Q(t) + R(t)$, for any time $t \geq 0$. The objective of the set of differential equations is to analyse the malware's transmission dynamics and the state of change of the sensor node.

The state transition equations are expressed as follows:

$$\begin{aligned}\frac{dS}{dt} &= \Lambda - \beta SI - (\mu + \theta)S \\ \frac{dE_1}{dt} &= a\beta SI + \sigma_1\beta VI - (\alpha_1 + \mu)E_1 \\ \frac{dE_2}{dt} &= b\beta SI + \sigma_2\beta VI - (\alpha_2 + \mu)E_2 \\ \frac{dV}{dt} &= \theta S - \sigma_1\beta VI - \sigma_2\beta VI - \mu V \\ \frac{dI}{dt} &= \alpha_1 E_1 + \alpha_2 E_2 - \mu I - \gamma I \\ \frac{dQ}{dt} &= \gamma I - \delta Q - \mu Q \\ \frac{dR}{dt} &= \delta Q - \mu R\end{aligned}\quad (5.1)$$

To model a dimensionally consistent model, we assume that the initial value of the seven compartments satisfy the initial conditions(ICs):

$$\begin{aligned}S(0) &= S_0, E_1(0) = E_{10}, E_2(0) = E_{20}, V(0) = V_0, I(0) \\ &= I_0, Q(0) = Q_0, R(0) = R_0,\end{aligned}\quad (5.2)$$

where $S_0, E_{10}, E_{20}, I_0, V_0, Q_0$ and R_0 are non-negative real numbers.

We want to turn the 5.1 model mentioned above into a fractional order problem in the corporate memory effect [55]. In order to do so, the equivalent integral form of (5.1) is provided by

$$\begin{aligned}S(t) &= S_0 + \int_0^t [\Lambda - \beta S(p)I(p) - (\mu + \theta)S(p)]dp \\ E_1(t) &= E_{10} + \int_0^t [a\beta S(p)I(p) + \sigma_1\beta V(p)I(p) \\ &\quad - (\alpha_1 + \mu)E_1(p)]dp \\ E_2(t) &= E_{20} + \int_0^t [b\beta S(p)I(p) + \sigma_2\beta V(p)I(p) \\ &\quad - (\alpha_2 + \mu)E_2(p)]dp\end{aligned}$$

$$\begin{aligned}V(t) &= V_0 + \int_0^t [\theta S(p) - \sigma_1\beta V(p)I(p) - \sigma_2\beta V(p)I(p) \\ &\quad - \mu V(p)]dp \\ I(t) &= I_0 + \int_0^t [\alpha_1 E_1(p) + \alpha_2 E_2(p) - \mu I(p) - \gamma I(p)]dp \\ Q(t) &= Q_0 + \int_0^t [\gamma I(p) - (\delta + \mu)Q(p)]dp \\ R(t) &= R_0 + \int_0^t [\delta Q(p) - \mu R(p)]dp\end{aligned}\quad (5.3a)$$

In order to include the influence of memory effect [49], we re-write (5.3a) in terms of time dependent integrals:

$$\begin{aligned}S(t) &= S_0 + \int_0^t \psi(t, p) [\Lambda - \beta S(p)I(p) - (\mu + \theta)S(p)]dp \\ E_1(t) &= E_{10} + \int_0^t \psi(t, p) [a\beta S(p)I(p) + \sigma_1\beta V(p)I(p) \\ &\quad - (\alpha_1 + \mu)E_1(p)]dp \\ E_2(t) &= E_{20} + \int_0^t \psi(t, p) [b\beta S(p)I(p) + \sigma_2\beta V(p)I(p) \\ &\quad - (\alpha_2 + \mu)E_2(p)]dp \\ V(t) &= V_0 + \int_0^t \psi(t, p) [\theta S(p) - \sigma_1\beta V(p)I(p) \\ &\quad - \sigma_2\beta V(p)I(p) - \mu V(p)]dp \\ I(t) &= I_0 + \int_0^t \psi(t, p) [\alpha_1 E_1(p) + \alpha_2 E_2(p) - \mu I(p) \\ &\quad - \gamma I(p)]dp \\ Q(t) &= Q_0 + \int_0^t \psi(t, p) [\gamma I(p) - (\delta + \mu)Q(p)]dp \\ R(t) &= R_0 + \int_0^t \psi(t, p) [\delta Q(p) - \mu R(p)]dp\end{aligned}\quad (5.3b)$$

Hence, in the conventional Markov process, $\psi(t, p)$ functions as a delta function $\delta(t, p)$, representing a time-dependent memory kernel. A power-law correlation function can be the appropriate choice for $\psi(t, p)$ in order to account for the long-term memory effect, which aids in the

system's evolution. Therefore, the non-local kernel can be assumed to be:

$$\psi(t, p) = \frac{(t-p)^{\alpha-1}}{\Gamma(\alpha)}, \alpha \in (0, 1].$$

Applying definition 2.1 to Eq. (5.3b) with this selection of $\psi(t, p)$, we

$$\begin{aligned} \text{obtain: } S(t) - S_0 &= {}_0^C D_t^{-\alpha} [\Lambda - \beta SI - (\mu + \theta)S] \\ E_1(t) - E_{10} &= {}_0^C D_t^{-\alpha} [a\beta SI + \sigma_1 \beta VI - (\alpha_1 + \mu)E_1] \\ E_2(t) - E_{20} &= {}_0^C D_t^{-\alpha} [b\beta SI + \sigma_2 \beta VI - (\alpha_2 + \mu)E_2] \\ V(t) - V_0 &= {}_0^C D_t^{-\alpha} [\theta S - \sigma_1 \beta VI - \sigma_2 \beta VI - \mu V] \\ I(t) - I_0 &= {}_0^C D_t^{-\alpha} [\alpha_1 E_1 + \alpha_2 E_2 - \mu I - \gamma I] \\ Q(t) - Q_0 &= {}_0^C D_t^{-\alpha} [\gamma I - \delta Q - \mu Q] \\ R(t) - R_0 &= {}_0^C D_t^{-\alpha} [\delta Q - \mu R] \end{aligned} \quad (5.4)$$

Using Lemma 1 and the Caputo fractional derivative of order α on both sides of each equation in (5.4), we can now obtain our fractional order epidemic model, which is determined by

$$\begin{aligned} {}_0^C D_t^\alpha S(t) &= \Lambda - \beta SI - (\mu + \theta)S \\ {}_0^C D_t^\alpha V(t) &= \theta S - \sigma_1 \beta VI - \sigma_2 \beta VI - \mu V \\ {}_0^C D_t^\alpha E_1(t) &= a\beta SI + \sigma_1 \beta VI - (\alpha_1 + \mu)E_1 \\ {}_0^C D_t^\alpha E_2(t) &= b\beta SI + \sigma_2 \beta VI - (\alpha_2 + \mu)E_2 \\ {}_0^C D_t^\alpha I(t) &= \alpha_1 E_1 + \alpha_2 E_2 - \mu I - \gamma I \\ {}_0^C D_t^\alpha Q(t) &= \gamma I - \delta Q - \mu Q \\ {}_0^C D_t^\alpha R(t) &= \delta Q - \mu R \end{aligned} \quad (5.5)$$

We will just concentrate on this model (5.5) because it is expected that the dimensions of both sides of the equations in (5.5) are the same.

Fractional derivatives are a very useful tool for describing memory effect when modelling the dynamics of malware propagation in WSNs. A smaller value of α indicates a slower decaying time-correlation function. For specificity, the fractional order α can be expressed as an index of memory of the system under consideration [49].

Description of Uniqueness and Existence of Non-Negative Solution of the Model

We need to prove the following theorem in order to address the existence and uniqueness of our model (5.5):

Theorem 6.1. For any $(S_0, E_{10}, E_{20}, V_0, I_0, Q_0, R_0) \in \mathbb{R}_+^7$, there exists a unique positive solution in the model (5.5)

with the initial conditions (ICs) (5.2). Additionally, the positively invariant compact set.

$$\Pi = \left\{ (S, E_1, E_2, V, I, Q, R) \in \mathbb{R}_+^7 : 0 \leq S + E_1 + E_2 + V + I + Q + R \leq \frac{\Lambda}{\mu} \right\} \quad (6.1)$$

contains all solutions to the model (5.5) that start in $\mathbb{R}_+^7$.

Proof: Let $X(t) = \begin{bmatrix} S(t) \\ E_1(t) \\ E_2(t) \\ V(t) \\ I(t) \\ Q(t) \\ R(t) \end{bmatrix}$; $X(0) = X_0 = \begin{bmatrix} S(0) \\ E_1(0) \\ E_2(0) \\ V(0) \\ I(0) \\ Q(0) \\ R(0) \end{bmatrix} =$

$$\begin{bmatrix} S_0 \\ E_{10} \\ E_{20} \\ V_0 \\ I_0 \\ Q_0 \\ R_0 \end{bmatrix} W(X(t)) = \begin{bmatrix} W_1(X(t)) \\ W_2(X(t)) \\ W_3(X(t)) \\ W_4(X(t)) \\ W_5(X(t)) \\ W_6(X(t)) \\ W_7(X(t)) \end{bmatrix} =$$

$$\begin{bmatrix} \Lambda - \beta SI - (\mu + \theta)S \\ a\beta SI + \sigma_1 \beta VI - (\alpha_1 + \mu)E_1 \\ b\beta SI + \sigma_2 \beta VI - (\alpha_2 + \mu)E_2 \\ \theta S - \sigma_1 \beta VI - \sigma_2 \beta VI - \mu V \\ \alpha_1 E_1 + \alpha_2 E_2 - \mu I - \gamma I \\ \gamma I - \delta Q - \mu Q \\ \delta Q - \mu R \end{bmatrix}$$

Therefore, the model (5.5) with the ICs (5.2), can be written in matrix form as:

$${}_0^C D_t^\alpha X(t) = W(X) \text{ with } X(0) = X_0.$$

The Jacobian matrix $J = \frac{\partial W}{\partial X} = \frac{\partial (W_1, W_2, W_3, W_4, W_5, W_6, W_7)}{\partial (S, E_1, E_2, V, I, Q, R)}$ of W is continuous on $\mathbb{R}_+^7$ by [[50], Remark 1.2.1] Since W is locally Lipschitz on $\mathbb{R}_+^7$ by [[51], Remark 3.8], there is a unique solution for every $(S_0, E_{10}, E_{20}, V_0, I_0, Q_0, R_0) \in \mathbb{R}_+^7$ in the model (5.5) with I.C.(5.2).

Now, We have to show that, for $(S_0, E_{10}, E_{20}, V_0, I_0, Q_0, R_0) \in \mathbb{R}_+^7$, the unique solution $(S, E_1, E_2, V, I, Q, R) \in \mathbb{R}_+^7$ of model (5.5) with the ICs (5.2), is non-negative. We refute this and talk about thirty distinct situations.

Case 1: There exist $t_S, t_{E_1}, t_{E_2}, t_I, t_V, t_R \in (0, \infty)$ such that.

$$1.1. S(t_S) < 0 \text{ and } S(t) \geq 0 \text{ for all } t \in [0, t_S) \quad (6.2)$$

$$1.2. E_1(t_{E_1}) < 0 \text{ and } E_1(t) \geq 0 \text{ for all } t \in [0, t_{E_1}) \quad (6.3)$$

$$1.3. E_2(t_{E_2}) < 0 \text{ and } E_2(t) \geq 0 \text{ for all } t \in [0, t_{E_2}) \quad (6.4)$$

$$1.4. V(t_V) < 0 \text{ and } V(t) \geq 0 \text{ for all } t \in [0, t_V) \quad (6.5)$$

$$1.5. I(t_I) < 0 \text{ and } I(t) \geq 0 \text{ for all } t \in [0, t_I) \quad (6.6)$$

$$1.6. Q(t_Q) < 0 \text{ and } Q(t) \geq 0 \text{ for all } t \in [0, t_Q] \quad (6.7)$$

$$1.7. R(t_R) < 0 \text{ and } R(t) \geq 0 \text{ for all } t \in [0, t_R] \quad (6.8)$$

Case 2: $S(t)$ is non-negative on $\mathbb{R}_+$ and 6.3, 6.4, 6.5, 6.6, 6.7 and 6.8 remains (as well as other comparable possible situations).

Case 3: $S(t), V(t)$ are non-negative on $\mathbb{R}_+$ and 6.3, 6.4, 6.6, 6.7 and 6.8 remains (as well as other comparable possible situations).

Case 4: $S(t), E_1(t), E_2(t), I(t)$ are non-negative on $\mathbb{R}_+$ and 6.5, 6.7 and 6.8 remains (as well as other comparable possible situations).

Case 5: $S(t), E_1(t), E_2(t), I(t), V(t)$ are non-negative on $\mathbb{R}_+$ and 6.7 and 6.8 remains (as well as other comparable possible situations).

Case 6: $S(t), E_1(t), E_2(t), I(t), V(t), Q(t)$ are non-negative on $\mathbb{R}_+$ and 6.8 remains (as well as other comparable possible situations).

We will demonstrate that none of the abovementioned situations can pass by employing the contradiction method.

Here are the seven scenarios that could occur if **Case 1** is true:

Subcase 1.1. For $t_S = \min\{t_S, t_{E_1}, t_{E_2}, t_V, t_I, t_Q, t_R\}$, from the first equation in (5.5), we have ${}_0^C D_t^\alpha S(t) \geq -\beta SI - (\mu + \theta)S \geq -k_1 S$ where, $k_1 = \beta \max_{t \in [0, t_S]} I(t) + \mu + \theta > 0$; thus, after solving, ${}_0^C D_t^\alpha S(t) \geq -k_1 S$ we get, $S(t) \geq S(0)\mathcal{E}_\alpha(-k_1 t^\alpha)$ for all $t \in [0, t_S]$. Hence, $S(t_S) \geq 0$. Which contradicts what we had assumed, $S(t_S) < 0$.

Subcase 1.2. For $t_{E_1} = \min\{t_S, t_{E_1}, t_{E_2}, t_V, t_I, t_Q, t_R\}$ from the second equation in (5.5), we have ${}_0^C D_t^\alpha E_1(t) \geq -(\alpha_1 + \mu)E_1 \geq -k_2 E_1$ where, $k_2 = (\alpha_1 + \mu) > 0$; thus, after solving, ${}_0^C D_t^\alpha E_1(t) \geq -k_2 E_1$ we get, $E_1(t) \geq E_1(0)\mathcal{E}_\alpha(-k_2 t^\alpha)$ for all $t \in [0, t_{E_1}]$. Hence, $E_1(t_{E_1}) \geq 0$. Which contradicts what we had assumed, $E_1(t_{E_1}) < 0$.

Subcase 1.3. For $t_{E_2} = \min\{t_S, t_{E_1}, t_{E_2}, t_V, t_I, t_Q, t_R\}$ from the third equation in (5.5), we have ${}_0^C D_t^\alpha E_2(t) \geq -(\alpha_2 + \mu)E_2 \geq -k_3 E_2$ where, $k_3 = (\alpha_2 + \mu) > 0$; thus, after solving, ${}_0^C D_t^\alpha E_2(t) \geq -k_3 E_2$ we get, $E_2(t) \geq E_2(0)\mathcal{E}_\alpha(-k_3 t^\alpha)$ for all $t \in [0, t_{E_2}]$. Hence, $E_2(t_{E_2}) \geq 0$. Which contradicts what we had assumed, $E_2(t_{E_2}) < 0$.

Subcase 1.4. For $t_V = \min\{t_S, t_{E_1}, t_{E_2}, t_V, t_I, t_Q, t_R\}$, from the fourth equation in (5.5), we have ${}_0^C D_t^\alpha V(t) \geq -((\sigma_1 + \sigma_2)\beta I + \mu)V \geq -k_4 V$ where, $k_4 = ((\sigma_1 + \sigma_2)\beta \max_{t \in [0, t_V]} I(t) + \mu) > 0$; thus, after solving ${}_0^C D_t^\alpha V(t) \geq -k_4 V$, we get, $V(t) \geq V(0)\mathcal{E}_\alpha(-k_4 t^\alpha)$ for all $t \in [0, t_V]$. Hence, $V(t_V) \geq 0$. Which contradicts what we had assumed, $V(t_V) < 0$.

Subcase 1.5. For $t_I = \min\{t_S, t_{E_1}, t_{E_2}, t_V, t_I, t_Q, t_R\}$, from the fifth equation in (5.5), we have ${}_0^C D_t^\alpha I(t) \geq -(\gamma + \mu)I \geq -k_5 I$ where, $k_5 = (\gamma + \mu) > 0$; thus, after solving ${}_0^C D_t^\alpha I(t) \geq -k_5 I$, we get, $I(t) \geq I(0)\mathcal{E}_\alpha(-k_5 t^\alpha)$ for all $t \in [0, t_I]$. Hence, $I(t_I) \geq 0$. Which contradicts what we had assumed, $I(t_I) < 0$.

Subcase 1.6. For $t_Q = \min\{t_S, t_{E_1}, t_{E_2}, t_V, t_I, t_Q, t_R\}$, from the sixth equation in (5.5), we have ${}_0^C D_t^\alpha Q(t) \geq -(\delta + \mu)Q \geq -k_6 Q$ where, $k_6 = (\delta + \mu) > 0$; thus, after solving ${}_0^C D_t^\alpha Q(t) \geq -k_6 Q$, we get, $Q(t) \geq Q(0)\mathcal{E}_\alpha(-k_6 t^\alpha)$ for all $t \in [0, t_Q]$. Hence, $Q(t_Q) \geq 0$. Which again contradicts what we had assumed, $Q(t_Q) < 0$.

Subcase 1.7. For $t_R = \min\{t_S, t_{E_1}, t_{E_2}, t_V, t_I, t_Q, t_R\}$, from the seventh equation in (5.5), we have ${}_0^C D_t^\alpha R(t) \geq \delta Q - \mu R \geq -k_7 R$ where, $k_7 = (\eta \max_{t \in [0, t_Q]} Q(t) + \mu) > 0$; thus, after solving ${}_0^C D_t^\alpha R(t) \geq -k_7 R$, we get, $R(t) \geq R(0)\mathcal{E}_\alpha(-t^\alpha)$ for all $t \in [0, t_R]$. Hence, $R(t_R) \geq 0$. Which will also contradict what we had assumed, $R(t_R) < 0$.

Thus, by adding together all of the subcases (1.1) to (1.7), we can conclude that the case (1) is not feasible. Assuming case (2) works now. Following the same procedure as in instance we can demonstrate the impossibility of case (2). Comparable reasoning can be employed to demonstrate that none of the cases—from case (2) to case (6) and beyond—occur, i.e., that all potential combinations of cases cannot occur. We came to the conclusion that the unique solution $(S, E_1, E_2, V, I, Q, R)$ is non-negative based on the analysis shown above. This result proves that our answer is non-negative. We now need to demonstrate the positive invariance of the set Π , as described by (6.1). The result of adding the equations in (5.5) is:

$$\begin{aligned} {}_0^C D_t^\alpha (S + E_1 + E_2 + V + I + Q + R) \\ = \Lambda - \mu S - \mu E_1 - \mu E_2 - \mu V - \mu I - \mu Q - \mu R \end{aligned}$$

for the sake of our convenience i.e. ${}_0^C D_t^\alpha M = \Lambda - \mu M$; where, $M = S + E_1 + E_2 + V + I + Q + R$

By resolving the equation [[22], remark 7.1], we obtain:

$$\begin{aligned} M(t) &= M(0)\mathcal{E}_\alpha(-\mu t^\alpha) + \Lambda \int_0^t p^{\alpha-1} \mathcal{E}'_\alpha(-\mu p^\alpha) dp \\ &= \left(-\frac{\Lambda}{\mu} + M(0)\right) \mathcal{E}_\alpha(-\mu t^\alpha) + \frac{\Lambda}{\mu} \end{aligned}$$

Hence, we note that $\mathcal{E}_\alpha(-\mu t^\alpha) \geq 0$. If $M(0) \leq \frac{\Lambda}{\mu}$, we have $S + E_1 + E_2 + V + I + Q + R = M(t) \leq \frac{\Lambda}{\mu}$. Hence, Φ is positively invariant. Finally, using Lemma 2, we were able to conclude that Υ encompasses all of the solutions for the model (5.5) which were first found in $\mathbb{R}_+^7$ as $\lim_{t \rightarrow \infty} \mathcal{E}_\alpha(-\mu t^\alpha) = 0$ we get $\lim_{t \rightarrow \infty} M(t) = \frac{\Lambda}{\mu}$. The proof is now complete.

Analyzing the Model's Equilibrium and Stability

Stability Analysis of Malware—Free Equilibrium

The unique malware free equilibrium of model (5.5) is

$$\omega_0 = \left(\frac{\Lambda}{\mu+\theta}, 0, 0, \frac{\theta\Lambda}{\mu(\mu+\theta)}, 0, 0, 0 \right)$$

$\therefore$ Jacobian corresponding to the model (5.5) is:

$$a_1 = (\alpha_1 + \alpha_2 + 3\mu + \gamma)$$

$$a_2 = [(\alpha_1 + \mu)(\alpha_2 + \mu) + (\gamma + \mu)(\alpha_1 + \alpha_2 + 2\mu)] - \frac{\beta\Lambda}{\mu(\mu + \theta)} [\mu(\alpha_1 a + \alpha_2 b) + \theta(\alpha_1 \sigma_1 + \alpha_2 \sigma_2)]$$

$$a_3 = [(\alpha_1 + \mu)(\alpha_2 + \mu)(\gamma + \mu)[1 - \mathfrak{R}_0]$$

It is clear that all coefficient of Eq. (7.4) are positive if $\mathfrak{R}_0 < 1$ and $\mu(\mu + \theta)[(\alpha_1 + \mu)(\alpha_2 + \mu) + (\gamma + \mu)((\alpha_1 + \alpha_2 + 2\mu)] - \beta\Lambda[\mu(\alpha_1 a + \alpha_2 b) + \theta(\alpha_1 \sigma_1 + \alpha_2 \sigma_2)] > 0$

$$J = \begin{pmatrix} -\beta I - \theta - \mu & 0 & 0 & 0 & -\beta S & 0 & 0 \\ a\beta I & -\mu - \alpha_1 & 0 & \beta I \sigma_1 & a\beta S + \beta V \sigma_1 & 0 & 0 \\ b\beta I & 0 & -\mu - \alpha_2 & \beta I \sigma_2 & b\beta S + \beta V \sigma_2 & 0 & 0 \\ \theta & 0 & 0 & -\mu - \beta I \sigma_1 - \beta I \sigma_2 & -\beta V \sigma_1 - \beta V \sigma_2 & 0 & 0 \\ 0 & \alpha_1 & \alpha_2 & 0 & -\gamma - \mu & 0 & 0 \\ 0 & 0 & 0 & 0 & \gamma & -\delta - \mu & 0 \\ 0 & 0 & 0 & 0 & 0 & \delta & -\mu \end{pmatrix} \quad (7.1)$$

At ω_0 , the Jacobian

Therefore, from Routh – Hurwitz criterion [53] all roots of Eq. (7.4) having negative real part, so system is asymptotically stable.

$$J(\omega_0) = \begin{pmatrix} -\theta - \mu & 0 & 0 & 0 & -\beta \frac{\Lambda}{\mu + \theta} & 0 & 0 \\ 0 & -\mu - \alpha_1 & 0 & 0 & (a\mu + \theta\sigma_1) \frac{\beta\Lambda}{\mu(\mu + \theta)} & 0 & 0 \\ 0 & 0 & -\mu - \alpha_2 & 0 & (b\mu + \theta\sigma_2) \frac{\beta\Lambda}{\mu(\mu + \theta)} & 0 & 0 \\ \theta & 0 & 0 & -\mu & \frac{-(\beta\sigma_1 + \beta\sigma_2)\theta\Lambda}{\mu(\mu + \theta)} & 0 & 0 \\ 0 & \alpha_1 & \alpha_2 & 0 & -\gamma - \mu & 0 & 0 \\ 0 & 0 & 0 & 0 & \gamma & -\delta - \mu & 0 \\ 0 & 0 & 0 & 0 & 0 & \delta & -\mu \end{pmatrix} \quad (7.2)$$

Its eigen values are

$$\lambda_1 = -(\theta + \mu), \lambda_2 = -\mu, \lambda_3 = -\mu, \lambda_4 = -(\delta + \mu) \quad (7.3)$$

And rest three eigen values are the roots of the cubic equation

$$a_0 \eta^3 + a_1 \eta^2 + a_2 \eta + a_3 = 0 \quad (7.4)$$

where $a_0 = 1$

here.

$$\mathfrak{R}_0 = \frac{\alpha_1(a\beta S_0 + \sigma_1 \beta V_0)}{(\alpha_1 + \mu)(\gamma + \mu)} + \frac{\alpha_2(b\beta S_0 + \sigma_2 \beta V_0)}{(\alpha_2 + \mu)(\gamma + \mu)} < 1,$$

where $\mathfrak{R}_0$ is dimensionless and is commonly known as basic reproduction number.

The following direct outcome can be obtained by analyzing Lemma 3 and above:

Table 1 Parameters used in model

S. No	Parameters	Explanation
1	Λ	Susceptible recruiting rate
2	θ	Rate of vaccination of vulnerable nodes
3	β	Infectious rate of Susceptible nodes
4	a, b	Rate of Susceptible nodes to become exposed
5	σ_1, σ_2	Rate of vaccinated nodes to become exposed
6	α_1, α_2	Infectious rate of an exposed state
7	γ	Quarantined rate of infected nodes
8	δ	Recovery rate of a quarantined node
9	μ	Mortality rate

Theorem 7.1. *The malware free equilibrium ω_0 is locally asymptotically stable if $\mathfrak{R}_0 < 1$ and unstable if $\mathfrak{R}_0 > 1$.*

According to the aforementioned theory, the unique solution of the model (5.5) with Initial Conditions (3.2) converges to ω_0 if the initial values $(S_0, E_{10}, E_{20}, V_0, I_0, Q_0, R_0) \in \mathbb{R}_+^7$ are sufficiently close to ω_0 . But here we are unable to illustrate the region of attraction and for this we talk about the following outcomes:

Theorem 7.2. *Take $\mathfrak{R}_0 < 1$. The region of attraction of the user-free equilibrium ω_0 is the compact set $\Pi = \{(S, E_1, E_2, V, I, Q, R) \in \mathbb{R}_+^7 : 0 \leq S + E_1 + E_2 + V + I + Q + R \leq \frac{\Lambda}{\mu}\}$.*

Proof. We must demonstrate that, for each $(S_0, E_{10}, E_{20}, V_0, I_0, Q_0, R_0) \in \Pi$, the unique solution $(S, E_1, E_2, V, I, Q, R)$ of model (5.5) with the Initial Conditions (ICs) (5.2) converges to ω_0 . Let the Lyapunov-function be.

$$L(t) = \frac{1}{2}I^2 + \frac{1}{2}\left(S - \frac{\Lambda}{\mu} + E_1 + E_2 + V + I + Q + R\right)^2 \quad (7.5)$$

By theorem 7.1, Π is non-negative invariant. Since, $(S_0, E_{10}, E_{20}, V_0, I_0, Q_0, R_0) \in \Pi$, we have $s \leq \frac{\Lambda}{\mu+\theta}$

Therefore, Lemma 4. and model (5.5), we get:

$$\begin{aligned} {}^C D_t^\alpha L &\leq I({}^C D_t^\alpha I(t)) \\ &+ \left(S - \frac{\Lambda}{\mu} + E_1 + E_2 + V + I + Q + R\right) {}^C D_t^\alpha \left(S - \frac{\Lambda}{\mu} + E_1 + E_2 + V + I + Q + R\right) \end{aligned}$$

$$\begin{aligned} &= I(\alpha_1 E_1 + \alpha_2 E_2 - \mu I - \gamma I) \\ &+ \left(S - \frac{\Lambda}{\mu} + E_1 + E_2 + V + I + Q + R\right) \\ &(\Lambda - \mu S - \mu E_1 - \mu E_2 - \mu V - \mu I - \mu Q - \mu R) \\ &\leq I(\alpha_1 E_1 + \alpha_2 E_2 - \mu I - \gamma I) \\ &+ \left(S - \frac{\Lambda}{\mu} + E_1 + E_2 + V + I + Q + R\right) \\ &(\Lambda - \mu(S + E_1 + E_2 + V + I + Q + R)) \\ &\leq I(\alpha_1 E_1 + \alpha_2 E_2 - \mu I - \gamma I) \\ &- \mu \left(S - \frac{\Lambda}{\mu} + E_1 + E_2 + V + I + Q + R\right)^2 \\ &\left(\because S \leq \frac{\Lambda}{\mu + \theta}\right). \end{aligned}$$

$${}^C D_t^\alpha L \leq 0 \text{ if and only if } I(\alpha_1 E_1 + \alpha_2 E_2 - \mu I - \gamma I) - \mu \left(S - \frac{\Lambda}{\mu} + E_1 + E_2 + V + I + Q + R\right)^2 \leq 0$$

$$\text{or } I(\alpha_1 E_1 + \alpha_2 E_2 - \mu I - \gamma I) \leq \mu \left(S - \frac{\Lambda}{\mu} + E_1 + E_2 + V + I + Q + R\right)^2$$

$$\text{Moreover, } {}^C D_t^\alpha L \leq 0 \text{ if and only if } S + E_1 + E_2 + V + I + Q + R = \frac{\Lambda}{\mu} \text{ and } I = 0.$$

$$\text{i.e. } \omega = \{(S, E_1, E_2, V, I, Q, R); {}^C D_t^\alpha L = 0\} = \{(S, E_1, E_2, V, I, Q, R); S + E_1 + E_2 + V + I + Q + R = \frac{\Lambda}{\mu}\}$$

Lemma 5., states that all solutions to model (5.5) that begin in Π tend to the largest invariant set in ω . So, for $I = 0, \lim_{t \rightarrow \infty} I(t) = 0$, using (5.5) we obtain

$$\begin{aligned} {}^C D_t^\alpha S(t) &= \Lambda - (\mu + \theta)S \\ {}^C D_t^\alpha E_1(t) &= -(\alpha_1 + \mu)E_1 \\ {}^C D_t^\alpha E_2(t) &= -(\alpha_2 + \mu)E_2 \\ {}^C D_t^\alpha V(t) &= \theta S - \mu V \\ {}^C D_t^\alpha Q(t) &= -\delta Q - \mu Q \\ {}^C D_t^\alpha R(t) &= \delta Q - \mu R \end{aligned} \quad (7.6)$$

Similar to how we obtained (5.5), we can also find the solution of (7.6), and after solving, we get, (similar to (5.5))

$$\begin{aligned} S(t) &= \left(-\frac{\Lambda}{(\mu + \theta)} + S(0)\right) \mathcal{E}_\alpha(-((\mu + \theta))t^\alpha) + \frac{\Lambda}{(\mu + \theta)} \\ E_1(t) &= E_1(0) \mathcal{E}_\alpha(-(\alpha_1 + \mu)t^\alpha) \\ E_2(t) &= E_2(0) \mathcal{E}_\alpha(-(\alpha_2 + \mu)t^\alpha) \\ V(t) &= \left(-\frac{S(t)\theta}{\mu} + V(0)\right) \mathcal{E}_\alpha(-\mu t^\alpha) + \frac{S(t)\theta}{\mu} \end{aligned}$$

$$Q(t) = Q(0)\mathcal{E}_\alpha(-(\delta + \mu)t^\alpha)$$

$$R(t) = \left(\frac{-\delta Q(t)}{\mu} + R(0)\right)\mathcal{E}_\alpha(-\mu t^\alpha) + \frac{Q(t)\delta}{\mu} \quad (7.7)$$

By Lemma 2.

$$\lim_{t \rightarrow \infty} S(t) = \frac{\Lambda}{(\mu + \theta)}, \quad \lim_{t \rightarrow \infty} E_1(t) = 0, \quad \lim_{t \rightarrow \infty} E_2(t) = 0,$$

$$\lim_{t \rightarrow \infty} V(t) = \frac{\theta}{\mu} \frac{\Lambda}{(\mu + \theta)}, \quad \lim_{t \rightarrow \infty} Q(t) = 0 \text{ and } \lim_{t \rightarrow \infty} R(t) = 0$$

Thus, $(S, E_1, E_2, V, I, Q, R) \rightarrow \omega_0$ as $t \rightarrow \infty$.

This completes the proof of the theorem.

Analyzing the Stability of Endemic Equilibrium

If $\mathfrak{R}_0 > 1$, Initially, we demonstrate that the model has a unique endemic equilibrium (5.5).

Theorem 7.3. If $\mathfrak{R}_0 > 1$. The model (5.5) has a unique endemic equilibrium $\omega^* = (S^*, E_1^*, E_2^*, V^*, I^*, Q^*, R^*)$, where $S^* = \frac{\Lambda\delta\gamma}{(\beta\mu(\delta + \mu)R^* + \delta\gamma(\mu + \theta))}$, $E_1^* = \frac{a\beta I^* S^* + \sigma_1 \beta V^* I^*}{\alpha_1 + \mu}$, $E_2^* = \frac{b\beta I^* S^* + \sigma_2 \beta V^* I^*}{\alpha_2 + \mu} = \frac{(\delta + \mu)\mu R^*}{\delta\gamma}$, $V^* = \frac{\theta S^*}{((\sigma_1 + \sigma_2)\beta I^* + \mu)}$, $Q^* = \frac{\mu R^*}{\delta}$ and R^* is the unique positive solution of the cubic equation in R .

Proof. Definition 2.3 states that the equations are satisfied at an endemic equilibrium point of the model (5.5).

$$0 = \Lambda - \beta S^* I^* - (\mu + \theta) S^*$$

$$0 = a\beta S^* I^* + \sigma_1 \beta V^* I^* - (\alpha_1 + \mu) E_1^*$$

$$0 = b\beta S^* I^* + \sigma_2 \beta V^* I^* - (\alpha_2 + \mu) E_2^*$$

$$0 = \theta S^* - \sigma_1 \beta V^* I^* - \sigma_2 \beta V^* I^* - \mu V^*$$

$$0 = \alpha_1 E_1^* + \alpha_2 E_2^* - \mu I^* - \gamma I^*$$

$$0 = \gamma I^* - \delta Q^* - \mu Q^*$$

$$0 = \delta Q^* - \mu R^* \quad (7.8)$$

Simple computation yields the endemic equilibrium points, which are given by

$$S^* = \frac{\Lambda\delta\gamma}{(\beta\mu(\delta + \mu)R^* + \delta\gamma(\mu + \theta))},$$

$$E_1^* = \frac{a\beta I^* S^* + \sigma_1 \beta V^* I^*}{(\alpha_1 + \mu)},$$

$$E_2^* = \frac{b\beta I^* S^* + \sigma_2 \beta V^* I^*}{(\alpha_2 + \mu)}, I^* = \frac{(\delta + \mu)\mu R^*}{\delta\gamma},$$

$$V^* = \frac{\theta S^*}{((\sigma_1 + \sigma_2)\beta I^* + \mu)}, Q^* = \frac{\mu R^*}{\delta},$$

where R^* is given by the equation

$$\Lambda - \frac{\Lambda\delta\gamma\mu}{(\beta\mu(\delta + \mu)R^* + \delta\gamma(\mu + \theta))} - \frac{a\mu\beta I^* S^* + \sigma_1 \mu\beta V^* I^*}{(\alpha_1 + \mu)}$$

$$- \frac{b\mu\beta I^* S^* + \sigma_2 \mu\beta V^* I^*}{(\alpha_2 + \mu)} - \frac{(\delta + \mu)\mu^2 R^*}{\delta\gamma}$$

$$- \frac{\mu\theta S^*}{((\sigma_1 + \sigma_2)\beta I^* + \mu)} - \frac{\mu^2 R^*}{\delta} - \mu R^*$$

$$= 0 \quad (7.9)$$

From Eq. (7.9) it's obvious that R^* have non-negative value if $\mathfrak{R}_0 > 1$. which proves the theorem (7.3).

Theorem 7.4. The endemic equilibrium $\omega^* = (S^*, E_1^*, E_2^*, V^*, I^*, Q^*, R^*)$ is globally asymptotically stable.

Proof: Let $\Phi(x) = x - x^* - x^* \ln \frac{x}{x^*}$

Define the Lyapunov function $L_1(t)$ by

$$L_1(t) = a_1 \Phi_1(S(t)) + a_2 \Phi_2(E_1(t))$$

$$+ a_3 \Phi_3(E_2(t)) + a_4 \Phi_4(V(t)) + a_5 \Phi_5(I(t))$$

$$+ a_6 \Phi_6(Q(t)) + a_7 \Phi_7(R(t)), \text{ where } a_i > 0, i = 1, 2, 3, 4, 5, 6, 7$$

$$L_1(t) = a_1 \left(S - S^* - S^* \ln \frac{S}{S^*} \right) + a_2 \left(E_1 - E_1^* - E_1^* \ln \frac{E_1}{E_1^*} \right)$$

$$+ a_3 \left(E_2 - E_2^* - E_2^* \ln \frac{E_2}{E_2^*} \right)$$

$$+ a_4 \left(V - V^* - V^* \ln \frac{V}{V^*} \right) + a_5 \left(I - I^* - I^* \ln \frac{I}{I^*} \right)$$

$$+ a_6 \left(Q - Q^* - Q^* \ln \frac{Q}{Q^*} \right)$$

$$+ a_7 \left(R - R^* - R^* \ln \frac{R}{R^*} \right) \quad (7.10)$$

With the help of lemma 4, the α -order Caputo derivatives of Φ_i will be calculated, $i = 1, 2, 3, 4, 5, 6$ and 7 by Lemma 4. The first equation of (7.8) gives

$$\Lambda = \beta S^* I^* + (\mu + \theta) S^*$$

Using this and the first equation of (5.5), we obtain

$${}_0^C D_t^\alpha \Phi_1(S) \leq \frac{S - S^*}{S} {}_0^C D_t^\alpha S(t)$$

$$= - \frac{(S - S^*)^2}{S} (\beta I^* + (\mu + \theta)) + \beta (S - S^*) (I^* - I) \quad (7.11)$$

From second equation of (7.8)

$$a\beta S^* I^* + \sigma_1 \beta V^* I^* = (\alpha_1 + \mu) E_1^*$$

Using this and the third equation in (5.5), we obtain

$$\begin{aligned}
{}_0^C D_t^\alpha \Phi_2(E_1(t)) &\leq \frac{(E_1 - E_1^*)}{E_1} {}_0^C D_t^\alpha E_1(t) \\
&= \frac{(E_1 - E_1^*)}{E_1} \left[a\beta \frac{(SIE_1^* - I^* S^* E_1)}{E_1^*} \right. \\
&\quad \left. + \sigma_1 \beta \frac{(VIE_1^* - I^* V^* E_1)}{E_1^*} \right] \quad (7.12)
\end{aligned}$$

From third equation of (7.8)

$$b\beta S^* I^* + \sigma_2 \beta V^* I^* = (\alpha_2 + \mu) E_2^*$$

Using this and the fourth equation of (5.5), we get

$$\begin{aligned}
{}_0^C D_t^\alpha \Phi_3(E_2(t)) &\leq \frac{(E_2 - E_2^*)}{E_2} {}_0^C D_t^\alpha E_2(t) \\
&= \frac{(E_2 - E_2^*)}{E_2} \left[b\beta \frac{(SIE_2^* - I^* S^* E_2)}{E_2^*} + \sigma_2 \beta \frac{(VIE_2^* - I^* V^* E_2)}{E_2^*} \right] \quad (7.13)
\end{aligned}$$

Now, from fifth equation of (7.8), we get

$$\alpha_1 E_1^* + \alpha_2 E_2^* = (\mu + \gamma) I^*$$

Using this and from fifth equation of (5.5), we get

$$\begin{aligned}
{}_0^C D_t^\alpha \Phi_4(I) &\leq \frac{(I - I^*)}{I} {}_0^C D_t^\alpha I(t) \\
&= \frac{(I - I^*)}{I} \left[\frac{\alpha_1 (E_1 I^* - E_1^* I)}{I^*} + \frac{\alpha_2 (E_2 I^* - E_2^* I)}{I^*} \right] \quad (7.14)
\end{aligned}$$

From fourth equation of (7.8), we get

$$(\sigma_1 + \sigma_2) \beta V^* I^* = \theta S^* - \mu V^*$$

Using this and from second equation of (5.5), we get

$$\begin{aligned}
{}_0^C D_t^\alpha \Phi_5(V) &\leq \frac{(V - V^*)}{V} {}_0^C D_t^\alpha V(t) \\
&= \frac{(V - V^*)}{V} \left(\frac{(SV^* I^* - S^* V I)}{V^* I^*} \theta + \frac{(I - I^*) \mu V}{I^*} \right) \quad (7.15)
\end{aligned}$$

From sixth equation of (7.8), we get

$$\gamma I^* = (\delta + \mu) Q^*$$

Using this and from sixth equation of (5.5), we get.

$$\begin{aligned}
{}_0^C D_t^\alpha \Phi_6(Q) &\leq \frac{(Q - Q^*)}{Q} {}_0^C D_t^\alpha Q(t) \\
&= \frac{(Q - Q^*)}{Q} \left(\frac{(IQ^* - \gamma I^* Q)}{Q^*} \right) \quad (7.16)
\end{aligned}$$

From seventh equation of (7.8), we get

$$\delta Q^* = \mu R^*$$

Using this and from seventh equation of (5.5), we get

$$\begin{aligned}
{}_0^C D_t^\alpha \Phi_7(R) &\leq \frac{(R - R^*)}{R} {}_0^C D_t^\alpha R(t) \\
&= \frac{(R - R^*)}{R} \left(\frac{(R^* Q - R Q^*) \delta}{R^*} \right) \quad (7.17)
\end{aligned}$$

From Eq. (7.11) to (7.17), it follows that ${}_0^C D_t^\alpha L_1(t) \leq 0$ and ${}_0^C D_t^\alpha L_1(t) = 0$ if and only if $S = S^*, E_1 = E_1^*, E_2 = E_2^*, I = I^*, Q = Q^*, V = V^*$ and $R = R^*$. Hence, the largest invariant set $\{(S, E_1, E_2, V, I, Q, R); {}_0^C D_t^\alpha L = 0\}$ is the singleton set $\{\omega^*\}$. based on theorem 5, we discovered that the set Π attracts all model (5.5) solutions that initiate in $\mathbb{R}_+^7$. According to Lemma 5, ω^* is globally asymptotically stable. Thus, the theorem is proved.

Analysis of the Numerical Solution

The considered model (5.1) has numerical solutions, which we may explore here. The suggested scheme is then used to obtain the numerical results. We approximate the Caputo fractional integral using the fractional Adams Bashforth method [54] in order to accomplish this goal.

The fundamental theorem of fractional calculus can be used to transform Eq. (5.1) into a fractional integral equation [22–24] as:

$$S(t) = S(0) + \frac{1}{\Gamma(\alpha)} \int_0^t (t-l)^{\alpha-1} \{\Lambda - \beta SI - (\mu + \theta)S\} dl$$

$$E_1(t) = E_1(0) + \frac{1}{\Gamma(\alpha)} \int_0^t (t-l)^{\alpha-1} \{a\beta SI + \sigma_1 \beta VI - (\alpha_1 + \mu)E_1\} dl$$

$$E_2(t) = E_2(0) + \frac{1}{\Gamma(\alpha)} \int_0^t (t-l)^{\alpha-1} \{b\beta SI + \sigma_2 \beta VI - (\alpha_2 + \mu)E_2\} dl$$

$$\begin{aligned}
V(t) &= V(0) \\
&\quad + \frac{1}{\Gamma(\alpha)} \int_0^t (t-l)^{\alpha-1} \{\theta S - \sigma_1 \beta VI - \sigma_2 \beta VI - \mu V\} dl
\end{aligned}$$

$$\begin{aligned}
I(t) &= I(0) \\
&\quad + \frac{1}{\Gamma(\alpha)} \int_0^t (t-l)^{\alpha-1} \{\alpha_1 E_1 + \alpha_2 E_2 - (\mu + \gamma)I\} dl
\end{aligned}$$

$$Q(t) = Q(0) + \frac{1}{\Gamma(\alpha)} \int_0^t (t-l)^{\alpha-1} \{\gamma I - (\delta + \mu)Q\} dl$$

$$R(t) = R(0) + \frac{1}{\Gamma(\alpha)} \int_0^t (t-l)^{\alpha-1} \{\delta Q - \mu R\} dl \quad (8.1)$$

To make things easier, let's assume

$$\begin{aligned}
H_1(t) &= \Lambda - \beta S(t)I(t) - (\mu + \theta)S(t), \\
H_2(t) &= a\beta S(t)I(t) + \sigma_1\beta V(t)I(t) - (\alpha_1 + \mu)E_1(t), \\
H_3(t) &= b\beta S(t)I(t) + \sigma_2\beta V(t)I(t) - (\alpha_2 + \mu)E_2(t), \\
H_4(t) &= \theta S(t) - \sigma_1\beta V(t)I(t) - \sigma_2\beta V(t)I(t) - \mu V(t), \\
H_5(t) &= \alpha_1 E_1(t) + \alpha_2 E_2(t) - \mu I(t) - \gamma I(t), \\
H_6(t) &= \gamma I(t) - \delta Q(t) - \mu Q(t), \\
H_7(t) &= \delta Q(t) - \mu R(t)
\end{aligned} \tag{8.2}$$

If and only if $S(t), E_1(t), E_2(t), V(t), I(t), Q(t)$ and $R(t)$ have an upper bound then only the expressions $H_1(t), H_2(t), H_3(t), H_4(t), H_5(t), H_6(t)$ and $H_7(t)$ are considered to satisfy the Lipschitz condition. If we consider two functions, $S_n(t)$ and $S_m(t)$, we obtain

$$\begin{aligned}
&\|H_{1n}(t) - H_{1m}(t)\| \\
&= \|-\beta(S_n - S_m)I - (\mu + \theta)(S_n - S_m)\| \\
&= \|-(\beta I + (\mu + \theta))(S_n - S_m)\| \\
&\leq \|-(\beta I + (\mu + \theta))\| \|S_n - S_m\| \\
&\leq \tau_1 \|S_n - S_m\|
\end{aligned} \tag{8.3}$$

where $\tau_1 = \|-(\beta \max_{t \in [0, t_3]} I(t) + (\mu + \theta))\|$.

Hence, we have

$$\|H_{1n}(t) - H_{1m}(t)\| \leq \tau_1 \|S_n - S_m\| \tag{8.4}$$

Similarly, we can obtain

$$\begin{aligned}
\|H_{2n}(t) - H_{2m}(t)\| &\leq \tau_2 \|E_{1n} - E_{1m}\| \\
\|H_{3n}(t) - H_{3m}(t)\| &\leq \tau_3 \|E_{2n} - E_{2m}\| \\
\|H_{4n}(t) - H_{4m}(t)\| &\leq \tau_4 \|V_n - V_m\| \\
\|H_{5n}(t) - H_{5m}(t)\| &\leq \tau_5 \|I_n - I_m\| \\
\|H_{6n}(t) - H_{6m}(t)\| &\leq \tau_6 \|Q_n - Q_m\| \\
\|H_{7n}(t) - H_{7m}(t)\| &\leq \tau_7 \|R_n - R_m\|
\end{aligned} \tag{8.5}$$

Consequently, all seven functions $H_1(t), H_2(t), H_3(t), H_4(t), H_5(t), H_6(t)$, and $H_7(t)$, satisfy the Lipschitz condition where $\tau_1, \tau_2, \tau_3, \tau_4, \tau_5, \tau_6$ and τ_7 are the corresponding Lipschitz constant.

For $p = 0, 1, 2, \dots$ the preceding equation is re-formulated as follows to begin an iterative scheme at given moment $t = t_{p+1}$

$$\begin{aligned}
S(t_{p+1}) &= S(0) + \frac{1}{\Gamma(\alpha)} \sum_{i=0}^{i=p} \int_{t_i}^{t_{i+1}} (t_{p+1} - l)^{\alpha-1} H_1(l) dl \\
E_1(t_{p+1}) &= E_1(0) + \frac{1}{\Gamma(\alpha)} \sum_{i=0}^{i=p} \int_{t_i}^{t_{i+1}} (t_{p+1} - l)^{\alpha-1} H_2(l) dl
\end{aligned}$$

$$\begin{aligned}
E_2(t_{p+1}) &= E_2(0) + \frac{1}{\Gamma(\alpha)} \sum_{i=0}^{i=p} \int_{t_i}^{t_{i+1}} (t_{p+1} - l)^{\alpha-1} H_3(l) dl \\
V(t_{p+1}) &= V(0) + \frac{1}{\Gamma(\alpha)} \sum_{i=0}^{i=p} \int_{t_i}^{t_{i+1}} (t_{p+1} - l)^{\alpha-1} H_4(l) dl \\
I(t_{p+1}) &= I(0) + \frac{1}{\Gamma(\alpha)} \sum_{i=0}^{i=p} \int_{t_i}^{t_{i+1}} (t_{p+1} - l)^{\alpha-1} H_5(l) dl \\
Q(t_{p+1}) &= Q(0) + \frac{1}{\Gamma(\alpha)} \sum_{i=0}^{i=p} \int_{t_i}^{t_{i+1}} (t_{p+1} - l)^{\alpha-1} H_6(l) dl \\
R(t_{p+1}) &= R(0) + \frac{1}{\Gamma(\alpha)} \sum_{i=0}^{i=p} \int_{t_i}^{t_{i+1}} (t_{p+1} - l)^{\alpha-1} H_7(l) dl
\end{aligned} \tag{8.6}$$

Within the interval $[t_i, t_{i+1}]$, the two points interpolation can be used to approximate the functions $H_1(l), H_2(l), H_3(l), H_4(l), H_5(l), H_6(l)$ and $H_7(l)$ inside the above integral.

$H_r(l) \cong \frac{H_r(t_i)}{h} (l - t_{i-1}) - \frac{H_r(t_{i-1})}{h} (l - t_i)$; where $r = 1, 2, 3, 4, 5, 6$, & 7 (8.7).

Using Eqs. (8.4) and (8.5) in (8.3) and on simplification we get:

$$\begin{aligned}
S(t_{p+1}) &= S(0) + \frac{1}{\Gamma(\alpha)} \sum_{i=0}^{i=p} \left(\frac{H_1(t_i)}{h} I_{i-1}^\alpha - \frac{H_1(t_{i-1})}{h} I_i^\alpha \right) \\
E_1(t_{p+1}) &= E_1(0) + \frac{1}{\Gamma(\alpha)} \sum_{i=0}^{i=p} \left(\frac{H_2(t_i)}{h} I_{i-1}^\alpha - \frac{H_2(t_{i-1})}{h} I_i^\alpha \right) \\
E_2(t_{p+1}) &= E_2(0) + \frac{1}{\Gamma(\alpha)} \sum_{i=0}^{i=p} \left(\frac{H_3(t_i)}{h} I_{i-1}^\alpha - \frac{H_3(t_{i-1})}{h} I_i^\alpha \right) \\
V(t_{p+1}) &= V(0) + \frac{1}{\Gamma(\alpha)} \sum_{i=0}^{i=p} \left(\frac{H_4(t_i)}{h} I_{i-1}^\alpha - \frac{H_4(t_{i-1})}{h} I_i^\alpha \right) \\
I(t_{p+1}) &= I(0) + \frac{1}{\Gamma(\alpha)} \sum_{i=0}^{i=p} \left(\frac{H_5(t_i)}{h} I_{i-1}^\alpha - \frac{H_5(t_{i-1})}{h} I_i^\alpha \right) \\
Q(t_{p+1}) &= Q(0) + \frac{1}{\Gamma(\alpha)} \sum_{i=0}^{i=p} \left(\frac{H_6(t_i)}{h} I_{i-1}^\alpha - \frac{H_6(t_{i-1})}{h} I_i^\alpha \right) \\
R(t_{p+1}) &= R(0) + \frac{1}{\Gamma(\alpha)} \sum_{i=0}^{i=p} \left(\frac{H_7(t_i)}{h} I_{i-1}^\alpha - \frac{H_7(t_{i-1})}{h} I_i^\alpha \right)
\end{aligned} \tag{8.8}$$

where,

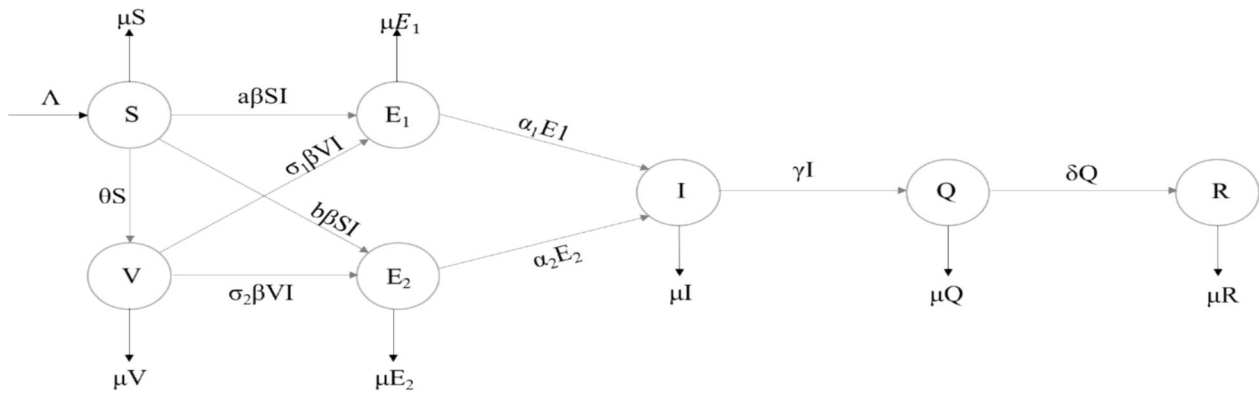


Fig. 1 Model diagram

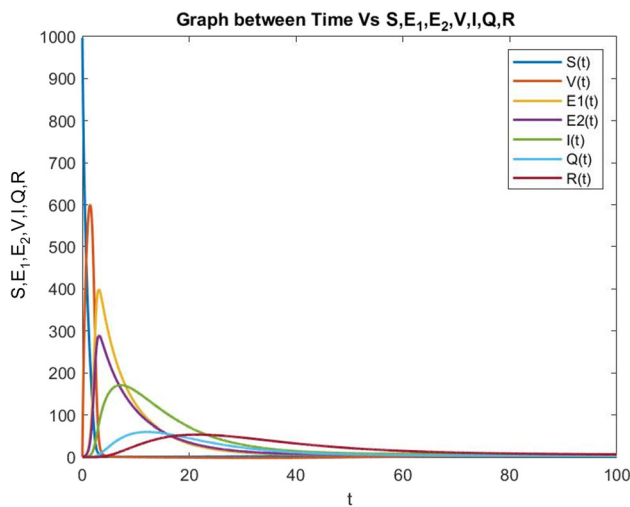


Fig. 2 Graph between time Vs S, E_1, E_2, V, I, Q, R

$$I_{i-1}^{\alpha} = \int_{t_i}^{t_{i+1}} (l - t_{i-1})(t_{p+1} - l)^{\alpha-1} dl$$

$$I_i^{\alpha} = \int_{t_i}^{t_{i+1}} (l - t_i)(t_{p+1} - l)^{\alpha-1} dl$$

On further simplification of the integrals I_{i-1}^{α} and I_i^{α} , and assuming, $t_i = ih$, we can easily find that

$$I_{i-1}^{\alpha} = \frac{h^{\alpha+1}}{\alpha(\alpha+1)} [(p+1-i)^{\alpha}(p-i+2+\alpha) - (p-i)^{\alpha}(p-i+2+2\alpha)] \quad (8.9)$$

$$I_i^{\alpha} = \frac{h^{\alpha+1}}{\alpha(\alpha+1)} [(p+1-i)^{\alpha+1} - (p-i)^{\alpha}(p-i+1+\alpha)] \quad (8.10)$$

Applying values from the Eqs. (8.9) and (8.10) in (8.8) and after some simplification, we obtain

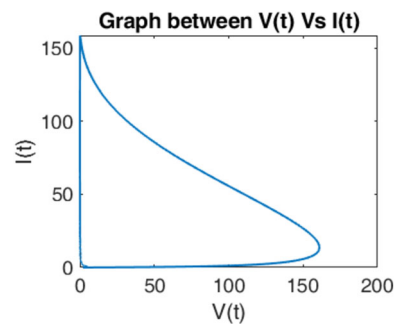


Fig. 3 Graph between $V(t)$ Vs $I(t)$

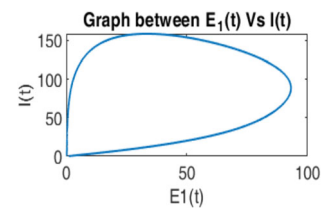


Fig. 4 Graph between E_1 Vs $I(t)$

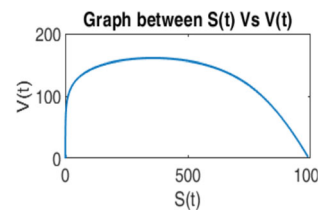


Fig. 5 Graph between $S(t)$ Vs $V(t)$

$$S(t_{p+1}) = S(0) + \sum_{i=0}^{p-1} \left(\frac{H_1(t_i)}{\Gamma(\alpha+2)} h^{\alpha} [(p+1-i)^{\alpha}(p-i+2+\alpha) - (p-i)^{\alpha}(p-i+2+2\alpha)] - \frac{H_1(t_{i-1})}{\Gamma(\alpha+2)} h^{\alpha} [(p+1-i)^{\alpha+1} - (p-i)^{\alpha}(p-i+1+\alpha)] \right)$$

$$\begin{aligned}
E_1(t_{p+1}) &= E_1(0) + \sum_{i=0}^{i=p} \left(\frac{H_2(t_i)}{\Gamma(\alpha+2)} h^\alpha [(p+1-i)^\alpha (p-i+2+\alpha) - (p-i)^\alpha (p-i+2+2\alpha)] \right. \\
&\quad \left. - \frac{H_2(t_{i-1})}{\Gamma(\alpha+2)} h^\alpha [(p+1-i)^{\alpha+1} - (p-i)^\alpha (p-i+1+\alpha)] \right) \\
E_2(t_{p+1}) &= E_2(0) + \sum_{i=0}^{i=p} \left(\frac{H_3(t_i)}{\Gamma(\alpha+2)} h^\alpha [(p+1-i)^\alpha (p-i+2+\alpha) - (p-i)^\alpha (p-i+2+2\alpha)] \right. \\
&\quad \left. - \frac{H_3(t_{i-1})}{\Gamma(\alpha+2)} h^\alpha [(p+1-i)^{\alpha+1} - (p-i)^\alpha (p-i+1+\alpha)] \right) \\
V(t_{p+1}) &= V(0) + \sum_{i=0}^{i=p} \left(\frac{H_4(t_i)}{\Gamma(\alpha+2)} h^\alpha [(p+1-i)^\alpha (p-i+2+\alpha) - (p-i)^\alpha (p-i+2+2\alpha)] \right. \\
&\quad \left. - \frac{H_4(t_{i-1})}{\Gamma(\alpha+2)} h^\alpha [(p+1-i)^{\alpha+1} - (p-i)^\alpha (p-i+1+\alpha)] \right) \\
(t_{p+1}) &= I(0) + \sum_{i=0}^{i=k} \left(\frac{H_5(t_i)}{\Gamma(\alpha+2)} h^\alpha [(p+1-i)^\alpha (p-i+2+\alpha) - (p-i)^\alpha (p-i+2+2\alpha)] \right. \\
&\quad \left. - \frac{H_5(t_{i-1})}{\Gamma(\alpha+2)} h^\alpha [(p+1-i)^{\alpha+1} - (p-i)^\alpha (p-i+1+\alpha)] \right) \\
Q(t_{p+1}) &= Q(0) + \sum_{i=0}^{i=p} \left(\frac{H_6(t_i)}{\Gamma(\alpha+2)} h^\alpha [(p+1-i)^\alpha (p-i+2+\alpha) - (p-i)^\alpha (p-i+2+2\alpha)] \right. \\
&\quad \left. - \frac{H_6(t_{i-1})}{\Gamma(\alpha+2)} h^\alpha [(p+1-i)^{\alpha+1} - (p-i)^\alpha (p-i+1+\alpha)] \right) \\
R(t_{p+1}) &= R(0) + \sum_{i=0}^{i=p} \left(\frac{H_7(t_i)}{\Gamma(\alpha+2)} h^\alpha [(p+1-i)^\alpha (p-i+2+\alpha) - (p-i)^\alpha (p-i+2+2\alpha)] \right. \\
&\quad \left. - \frac{H_7(t_{i-1})}{\Gamma(\alpha+2)} h^\alpha [(p+1-i)^{\alpha+1} - (p-i)^\alpha (p-i+1+\alpha)] \right)
\end{aligned} \tag{8.11}$$

Stability Analysis of the Numerical Solution Obtained Through Fractional Adams Bashforth Method [55]:

This is clear from (8.4) and (8.5) that the functions expressions $H_1(t), H_2(t), H_3(t), H_4(t), H_5(t), H_6(t)$ and $H_7(t)$ are satisfying the Lipschitz condition,

Now,

$$\begin{aligned}
&\|H_1(t_p) - H_1(t_{p-1})\|_\infty \\
&= \|-\beta S(t_p)I(t_p) - (\delta_1 + \gamma)S(t_p) + \beta S(t_{p-1})I(t_{p-1}) + (\delta_1 + \gamma)S(t_{p-1})\|_\infty \\
&= \|-\beta S(t_p)(I(t_p) - I(t_{p-1})) \\
&\quad - \beta(S(t_{p-1}) - S(t_p))I(t_{p-1}) - (\delta_1 + \gamma)(S(t_p) - S(t_{p-1}))\|_\infty
\end{aligned}$$

$$\begin{aligned}
&\leq \|-\beta S(t_p)\|_\infty \|I(t_p) - I(t_{p-1})\|_\infty \\
&\quad + \|S(t_p) - S(t_{p-1})\|_\infty \|\beta I(t_{p-1})\|_\infty \\
&\quad + \|-(\delta_1 + \gamma)\|_\infty \|S(t_p) - S(t_{p-1})\|_\infty
\end{aligned} \tag{8.12}$$

using (8.11) in (8.12) and from [55], it is quite obvious.

as $p \rightarrow \infty$, $\|S(t_p) - S(t_{p-1})\|_\infty \rightarrow 0$ and $\|I(t_p) - I(t_{p-1})\|_\infty \rightarrow 0$ (8.13).

Hence, as $p \rightarrow \infty$, $\|H_1(t_p) - H_1(t_{p-1})\|_\infty \rightarrow 0$ (using (8.11) in (8.12)), so, Theorem 3.3 shows that the solution $S(t_p)$ is stable.

In a similar manner, the stability condition for solutions can be determined.

$E_1(t_p), E_2(t_p), V(t_p), I(t_p), Q(t_p)$ and $R(t_p)$.

Discussion

Figure 2 illustrates the time-dependent dynamics of various classes in the system, including the susceptible (S), vaccinated (V), exposed (E_1 & E_2), infected (I), quarantined (Q), and recovered (R) classes. Our analysis reveals that the system is asymptotically stable. Notably, we observed that while the number of worms (malware) initially increases, it gradually diminishes over time once the reproduction number $\mathcal{R}_0$ drops below one.

Figure 3 demonstrates the impact of vaccination on malware propagation in Wireless Sensor Networks (WSNs). By comparing the vaccination rate to the infection rate, it is evident that as the vaccination rate increases, the number of infected nodes decreases. This suggests that node vaccination plays a significant role in enhancing the stability and extending the lifetime of WSNs.

In Fig. 4, a comparison between the number of infected and exposed nodes is presented, highlighting the dynamics of malware spread within the network. Finally, Fig. 5 compares the susceptible and vaccination rates, showing that immunization effectively curtails malware propagation by converting susceptible nodes into vaccinated ones. This highlights the potential of vaccination as a strategic intervention to control malware outbreaks in WSNs.

Conclusion

Fractional-order derivatives are ideal for modeling real-world phenomena because they give a wider range to choose the derivative order. Its flexibility makes it better at fitting real-time data than integer-order models. We propose a fractional-order $S E_1 E_2 VIQR$ epidemic model to prevent malware outbreaks and prolong Wireless Sensor Networks (WSN) lifespan. This model allows for faster detection of sensor nodes changing from susceptible to infected, allowing for timely actions to preserve node functionality and stabilize the WSN.

We attempt to comprehend the dynamics of malware spread in WSN and the memory impact in the epidemiological modeling with the use of fractional differential equations. Basic reproduction number ($\mathcal{R}_0$) has been calculated to analyse the critical constraint system dynamics. This parameter uses to investigate the system stability conditions. From various observations it is established that on the one hand when the value of $\mathcal{R}_0 < 1$, WSN will be stabilize in malware-free state and on the other hand if $\mathcal{R}_0 > 1$ WSN will exist in the endemic state. The stability of the network is discussed in different conditions and consequently their theorem and its proof have been conferred. The recovery method used to overcome the problem

of malware attack has also been analysed. Further study is needed to optimize these models to improve strategies for control and reduce epidemic impact in WSNs. Our future research will examine how variable-order fractional derivative models and artificial intelligence might improve epidemic management in these networks.

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Data Availability No data were used to support this study.

Declarations

Conflicts of Interest The authors declare that they have no conflicts of interest.

Research Involving Human and / or Animals No animal involves only Author being human involve.

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